

Relationship between spectral induced polarization and hydraulic properties of saturated and unsaturated sandstone

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Received 21 April 2005; revised 29 July 2005; accepted 16 August 2005; published 14 December 2005.

[1] There is growing interest in the use of geophysical methods for hydrological model parameterization. Empirical induced polarization (IP)–hydraulic conductivity (K) relationships have been developed, but these are only applicable to sediments in which the IP response shows limited variation with electrical current frequency. Here we examine the spectral IP response of samples taken from a UK sandstone aquifer and compare measured parameters with physical and hydraulic properties. We demonstrate the limited value of existing IP- K models due to the inherent IP frequency dependence of these samples. Our results show how the mean relaxation time, τ , is a more appropriate measure of IP response for these sediments. A significant inverse correlation between the surface area to pore volume ratio and τ is observed, suggesting that τ is a measure of a characteristic hydraulic length scale. This is supported by a measured strong positive correlation between $\log \tau$ and $\log K$. Our measurements also reveal evidence of a relationship between τ and a dominant pore throat size, which leads to postulations about the parallelism between the spectral IP behavior and unsaturated hydraulic characteristics. Additional experiments show how the relaxation time is affected by degree of fluid saturation, indicating that saturation levels must be accounted for if our empirical relationships are applied to vadose zone studies. Our results show clear evidence of the potential value of frequency-based IP measurements for parameterization of groundwater flow models.

Citation: Binley, A., L. D. Slater, M. Fukes, and G. Cassiani (2005), Relationship between spectral induced polarization and hydraulic properties of saturated and unsaturated sandstone, *Water Resour. Res.*, 41, W12417, doi:10.1029/2005WR004202.

1. Introduction

[2] Accurate modeling of groundwater flow and solute transport is essential to make reliable assessments of aquifer resources vulnerability. However, the characterization of flow and transport properties of soils and rocks is a challenging task, particularly in unsaturated states, due to the lack of suitable in situ measurement techniques. Laboratory measurements can provide valuable information on the soil characteristics at sampling locations, however, sampling can never be carried out with sufficient spatial density to allow a thorough site characterization. In addition, sample disturbance is difficult to avoid. Consequently, there is a strong need for reliable, spatially extensive and minimally invasive methods that can measure physical properties related to the hydrologic quantities of interest. Geophysical exploration techniques have been increasingly used to infer the properties of the saturated and the unsaturated zones. Besides the advantage of minimal disturbance

to the subsurface, geophysical methods have an additional, significant, advantage over conventional hydrological measurements in that they provide measurements over relatively large support volumes.

[3] Electromagnetic (ground) conductivity, resistivity and ground penetrating radar (GPR) geophysical methods have been widely used in hydrological studies to infer lithological boundaries and help with the development of conceptual models of the subsurface. Relationships between geophysical parameters and hydraulic properties have been widely investigated. In saturated soils the dependence of the dielectric constant on porosity, for example, is well established [Birchak *et al.*, 1974]. Similarly, porosity-electrical conductivity (σ) relationships, through Archie's law [Archie, 1942], for example, are well known. In unsaturated soils, the variation in dielectric constant with moisture content [e.g., Topp *et al.*, 1980] has been universally accepted and forms the basis behind the use of time domain reflectometry (TDR) for moisture content estimation [Robinson *et al.*, 2003]. Archie type relationships between soil water content and σ have also been used to infer moisture content variation in the subsurface using electrical and electromagnetic methods. Relationships between σ and hydraulic conductivity (K) have been explored in a range of subsurface environments. In clay rich porous soils the electrical conductivity will increase as clay content increases and through this a negative correlation between electrical conductivity and hydraulic conductivity may exist

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[e.g., *Kosinski and Kelly*, 1981]. In clay free soils a positive correlation between σ and K may be apparent as the electrical conductivity will be strongly (and positively) influenced by porosity (Φ). Furthermore, the much cited analogy between groundwater flow and electrical flow is limited by the fact that electrical conductivity scales with Φ to the power 1.5–2.5 in loosely consolidated materials whereas hydraulic conductivity scales with Φ to a much higher order. More significant, however, is that hydraulic conductivity is principally controlled by an effective pore size whereas electrical conductivity is a function of porosity. It would therefore appear that universal models explaining the relationship between σ and K are unlikely to exist [*Purvanec and Andricevic*, 2000].

[4] These relationships are all based on the sensitivity of geophysical parameters to the relative volumes of the pore, liquid and gas phases of the medium. The saturated hydraulic conductivity is, however, strongly influenced by geometrical and physical characteristics of the internal surfaces of the porous medium in addition to bulk volume measures. Geophysical techniques that depend on surface characteristics are scarce. One such method, induced polarization (IP), which has been used for a number of years for other purposes (mainly mineral exploration) [*Sumner*, 1976] has received significant recent attention for application to hydrological characterization and it is the possible link between IP and hydraulic properties of sediments that is the focus of this study.

1.1. Induced Polarization

[5] If a constant electrical current in the ground is interrupted, the voltage across the potential electrodes does not drop to zero instantaneously. It is found to relax for a time ranging from seconds to minutes, starting from an initial value that is a small fraction of the voltage that existed when the current was flowing. An analogous behavior can be observed in the frequency domain, where a phase shift between injected current and voltage drop can be measured. The IP effect can therefore be described in terms of complex electrical resistivity, ρ^* , or complex electrical conductivity, σ^* :

$$\sigma^*(\omega) = \frac{1}{\rho^*(\omega)} = \sigma'(\omega) + i\sigma''(\omega), \quad (1)$$

where σ' and σ'' are the measured real and imaginary components of electrical conductivity, respectively, ω is the excitation frequency and $i = \sqrt{-1}$. The real part of σ^* represents the total energy loss (conduction) whereas the imaginary part represents the total storage loss (polarization). The investigation of complex conductivity dependence on ω is often referred to as spectral induced polarization (SIP). The measurements are usually recorded as a conductivity ($|\sigma|$) or resistivity ($|\rho|$) magnitude and a phase angle, ϕ , both being related to the conduction and storage mechanisms:

$$|\sigma| = \frac{1}{|\rho|} = \sqrt{\sigma'^2 + \sigma''^2}$$

$$\phi = \tan^{-1} \left(\frac{\sigma''}{\sigma'} \right). \quad (2)$$

The mechanisms of charge storage vary with excitation frequency (see *Olhoeft* [1985] for a review). In the low-

frequency range of SIP (<1000 Hz) the dominant mechanism is the polarization of the ionic charge associated with the electrical double layer that exists at the mineral-fluid interface.

[6] Previous work shows that the magnitude of this polarization at a single measurement frequency (i.e., σ'') is closely related to microgeometrical properties. It is primarily controlled by the total mineral-fluid surface area within the sample, which can be estimated from the total mineral surface area per unit pore volume (S_{por}). Previous work has shown σ'' to exhibit a near linear dependence on S_{por} [*Börner and Schön*, 1991; *Börner et al.*, 1996; *Slater and Glaser*, 2003]. As grain size and surface are inversely related, other workers have found a near-inverse linear relation between σ'' and effective grain size [*Vanhala*, 1997a; *Slater and Lesmes*, 2002] although the exact nature of this relationship depends on how the surface area is distributed across the range of grain sizes [*Slater and Glaser*, 2003]. Models linking pore size, rather than grain size, to this polarization have also been proposed [e.g., *Titov et al.*, 2002]. Fluid chemistry and saturation also influence the polarization and may complicate interpretation of microgeometrical properties [*Klein and Sill*, 1982; *Vinegar and Waxman*, 1984; *Vanhala et al.*, 1992].

[7] Important microgeometrical information is also potentially obtainable by analyzing the nature of the frequency dependence of $\sigma^*(\omega)$. This results from the fact that the relaxation time is related to the length scale of the induced charge separation along the mineral-fluid interface. *Scott and Barker* [2003] recently studied the relationship between pore throat size in sandstones and the frequency-dependent polarization. Their investigation considered a range of Permo-Triassic sandstones obtained from a number of sites across the UK. For these sandstones *Scott and Barker* [2003] showed a convincing correlation between a “dominant pore throat” diameter and the frequency at which a peak in the phase angle occurs. If such relationships exist then one may also hypothesize a link between hydraulic characteristics and IP spectra.

1.2. SIP Models

[8] Currently there are no universal physically based models which describe the frequency-dependent complex conductivity response of sediments and consequently macroscopic representations have been adopted in the literature. A review of existing approaches is given by *Dias* [2000]. One of the most popular among such relaxation models is the Cole-Cole (C-C) model [*Pelton et al.*, 1978]. This four parameter model describes SIP in terms of a complex electrical conductivity (or electrical resistivity) function, showing variations of its real and imaginary part with excitation frequency. In terms of complex conductivity [*Jones*, 2002],

$$\sigma^*(\omega) = \sigma_0 \left[1 + m \left(\frac{(i\omega\tau)^c}{1 + (i\omega\tau)^c(1-m)} \right) \right], \quad (3)$$

where σ_0 is the conductivity at DC (very low) frequency, τ is the mean relaxation time, c is an exponent that typically takes values in the range 0.1–0.6 and m is the chargeability (given by $m = 1 - \sigma_0/\sigma_\infty$, where σ_∞ is the conductivity at high frequency).

[9] Spectra of the Cole-Cole type show a maximum phase angle at the frequency [Major and Silic, 1981]:

$$\omega_{peak} = \frac{1}{\tau(1-m)^{c/2}}. \quad (4)$$

Spectral induced polarization measurements may not reveal such variation in phase angle and, under these circumstances, the simpler two parameter constant phase angle (CPA) model can be adopted. The CPA model may be expressed as [Lesmes and Friedman, 2005]

$$\sigma^*(\omega) = \sigma_n \left(\frac{i\omega}{\omega_0} \right)^p, \quad (5)$$

where σ_n is the magnitude of the complex conductivity measured at some arbitrary frequency (ω_0 , typically equivalent to 1 Hz) and p is a parameter describing a power law dispersion in the spectral response.

1.3. Hydraulic-Electrical Polarization Relationships

[10] Recognizing the importance of surface area on fluid flow, a number of workers have investigated the possibility of using IP to estimate the hydraulic conductivity of a range of sediments. *Worthington and Collar* [1984] and *Olorunfemi and Griffiths* [1985] discuss the potential link between IP and hydraulic conductivity of sandstones and demonstrate evidence of relationships, using chargeability as a measure of IP. *Olorunfemi and Griffiths* [1985] fail to account for the effect of sample electrical conductivity on chargeability which may account for their observed apparent increase in IP effect with increase in hydraulic conductivity.

[11] Using the relationship between the IP effect and the ratio of the surface area of pores to the volume of pores (S_{por}) demonstrated by *Börner and Schön* [1991], *Börner et al.* [1996] proposed a model relating electrical parameters to saturated hydraulic conductivity. Their approach, based on a modified Kozeny-Carman model, results in the following equation for hydraulic conductivity (K):

$$K = \frac{a}{FS_{por}^b}, \quad (6)$$

where a and b are empirical constants and F is the formation factor (the ratio of fluid electrical conductivity to bulk electrical conductivity, ignoring surface conduction effects). Application of the fractal based PARIS model of *Pape et al.* [1982] to a range of rock types results in estimates of a and b as 641 and 3.1, respectively (K in m d^{-1} , S_{por} in μm^{-1}). *Börner et al.* [1996] used IP measurements to determine estimates of S_{por} in unconsolidated sands and gravels, and from these coupled with measurements of K , observed b in equation (6) to lie in the range 2.8 to 4.6.

[12] The link between the magnitude of the IP effect and S_{por} shown by *Börner and Schön* [1991] is based on an assumed CPA model (equation (5)) describing the frequency dependence of complex electrical conductivity. *Börner and Schön* [1991] showed, empirically, a linear relationship between $\sigma''(\omega_0)$ and S_{por} and *Börner et al.* [1996] demonstrated field application of this approach to unconsolidated sands and gravels with some success.

[13] *de Lima and Niwas* [2000] also adopted a Kozeny-Carman type of petrophysical model linking IP to hydraulic conductivity in shaly sandstones. They defined a “lithoporosity factor” which accounts for the clay content, porosity and hydraulic tortuosity through parameters derived from geoelectrical measurements. An equivalent volume electrical conductivity for the shale layer coating the surface of pores was determined from the σ'' measurements. Frequency dependence of the IP response is ignored in the *de Lima and Niwas* [2000] approach; instead, they formulate their model in terms of conventional time domain IP parameters.

[14] *Slater and Lesmes* [2002] reviewed the Börner model in their study of laboratory based measurements using natural sands, glacial tills and artificial sand-clay mixes. On the basis of electrical measurements on a set of samples with five orders of magnitude range in hydraulic conductivity they revealed limitations in practical application of the Börner model and proposed a simpler model using grain size (an easier measurement to obtain than S_{por} for model calibration purposes), based on a Hazen-type empirical model, rather than surface area, as the common link between the imaginary component of electrical conductivity and hydraulic conductivity.

[15] While these models show encouraging success for samples displaying limited frequency dependence of the phase angle, it is unclear how universally they may be applied. Consolidated rocks may display a distinct peak in the phase of the conductivity spectra [e.g., *Scott and Barker*, 2003; *Clennell and de Lima*, 2003] such that the CPA model is invalid. In such cases it appears inappropriate to adopt a single frequency imaginary conductivity as a representative measure of polarization and then relate this to surface area or some other geometrical factor influencing hydraulic conductivity. In these instances it is necessary to fit a C-C type model that accounts for the phase peak in the dispersion. The mean relaxation time, τ , in the C-C model, has been shown to be influenced by grain size [e.g., *Vanhala*, 1997a] and the recent work of *Scott and Barker* [2003] demonstrated a strong link between a characteristic pore size and a measure of relaxation time (although they did not specifically address relaxation models in their study). It is logical to propose therefore that for sediments displaying significant frequency dependence of polarization, the relaxation time of a C-C model may offer a useful insight into factors controlling the hydraulic conductivity.

[16] If such relationships are observed then opportunities may exist to determine useful hydraulic characteristics in situ from frequency-dependent electrical measurements. Furthermore, the development of subsurface imaging techniques for IP investigations [e.g., *Kemna et al.*, 2004] may permit detailed characterization of the spatial variability of these hydraulic properties, with obvious potential for hydrological model parameterization. However, it is recognized that field deployment of these techniques is generally limited to a narrow (low) frequency range (based on our experience, less than 10 Hz with currently available field instrumentation) due to capacitive and electromagnetic coupling effects.

[17] To determine the practical value of any observed electrical-hydraulic relationship it is also necessary to examine the effect of changes in water saturation on the measured IP response (since it is expected that field inves-

tigations of vadose zone hydrological model parameters are likely to be conducted under unsaturated soil water conditions). Few investigations of such effects have been reported. The dependence of higher-frequency polarization on variably saturated sandstones was reported by *Knight and Nur* [1987] but these were limited to a 10 kHz to 1 MHz frequency range. *Su et al.* [2000] also examined polarization of sandstones under varying water saturation but restricted their study to 100 Hz to 15 MHz frequencies. More recently, *Ulrich and Slater* [2004] reported measurements of the SIP response in the 0.1 Hz to 1 kHz frequency range on unconsolidated sand samples under variable water saturation. The power law increase in σ'' with saturation as observed by *Ulrich and Slater* [2004] suggests that polarization in soils may be enhanced by zones of reduced ionic mobility forming at pore throats [*Schön*, 1996; *Ulrich and Slater*, 2004]. Such zones act as ion selective membranes and increase the effective polarization of the sample. Although the samples of *Ulrich and Slater* [2004] showed only limited frequency dependence, a shift in phase peak to higher frequencies with reducing saturation is consistent with the conceptual model of early drainage of the larger pore throats. *Titov et al.* [2004], albeit on the basis of a limited set of measurements, suggest a similar dependence of polarization on water saturation, which supports other findings [for example, *Ogilvy and Kuzmina*, 1972; *Iliceto et al.*, 1982; *Olorunfemi and Griffiths*, 1985]. Clearly, these findings may limit the usefulness of using SIP measurements for determination of hydraulic conductivity in the vadose zone.

1.4. Aims

[18] Our general aims are to investigate the link between hydraulic and electrical properties of sediments within a specific UK Triassic sandstone aquifer. We wish to assess the potential value of using IP for constraining parameters of vadose zone models which are necessary for predicting recharge processes within the aquifer. To achieve this we elected to carry out measurements on samples extracted from this aquifer at two sites where significant previous work had been carried out. The two sites, Eggborough and Hatfield (for location, see Figure 1), have been studied as part of an ongoing program to assess aquifer vulnerability using hydrogeophysical methods. *Binley et al.* [2001] have shown at the Hatfield site how cross borehole radar can be effectively used to monitor the movement of the wetting front in the sandstone under natural recharge. Using borehole arrays, *Binley et al.* [2002a] have mapped water tracer migration in the vadose zone using resistivity and radar methods and used the results to infer effective hydraulic properties in vadose zone models. *Binley et al.* [2002b], using radar and resistivity, show how seasonal variation in moisture profiles at the two sites show significant similarity, the results from which have been used to develop models of natural recharge processes [*Binley and Beven*, 2002; *Cassiani and Binley*, 2005]. *Brovelli et al.* [2005] examined the relationship between electrical properties (DC electrical conductivity and electrical permittivity) and saturation in sandstones from the sites, using pore-scale modeling tools. In addition, *West et al.* [2003] have developed petrophysical models for local sandstones in order to link volumetric moisture content to high-frequency (radar) electrical properties. Here, rather than exploring potential universal rela-



Figure 1. Location of field sites. Hatfield site: 53.56365° latitude, −1.01557° longitude. Eggborough site: 53.70213° latitude, −1.13807° longitude.

tionships between electrical and hydraulic properties, we wish to determine whether useful relationships exist that can be applied for hydrological model development for this specific aquifer.

[19] A specific aim of our work is to evaluate the usefulness of the model of *Börner et al.* [1996] in relating IP and hydraulic conductivity in a range of lithologies at the two sites. Previous spectral IP measurements on UK Triassic sandstones have indicated significant frequency dependence of polarization [*Scott and Barker*, 2003; *Clennell and de Lima*, 2003] and therefore we hypothesize limited value of the *Börner et al.* [1996] model. We therefore also wish to explore the hydraulic information gained from spectral IP measurements on these sediments with a view to determining a more appropriate measure of polarization that relates to hydraulic conductivity.

[20] Given that our ongoing aim is to utilize geophysics in parameterization of vadose zone flow models applied to this important UK aquifer, it is necessary to evaluate the effect of partial fluid saturation on IP measures. In addition, recognizing that a broad frequency range of electrical measurements is somewhat difficult to obtain in the field we also wish to examine the usefulness of field measurable properties in predicting hydraulic properties (as studied by *Slater and Lesmes* [2002] on unconsolidated sand/clay mixes). To achieve these aims a series of experimental measurements of physical and electrical properties were performed on samples extracted from the two UK sites.

2. Methodology

[21] The Sherwood Sandstone at the two field sites is fluvially derived and is mainly medium and fine grained,

although occasional siltstone bands are observed. At both sites drift cover is minimal (typically 2 to 3 m thick) and the water table is typically 17 m (Eggborough) and 10 m (Hatfield) deep. In March 2003 a 20 m long, 100 mm diameter drill core was extracted from the Eggborough site. To supplement the core, blocks of sandstone (typically 200 mm $\times$ 200 mm $\times$ 200 mm) were obtained from quarries at Eggborough and Hatfield. Care was taken to ensure that the quarry samples were not extensively weathered by removing at least 300 mm sandstone prior to excavation. In what follows we refer to these quarry samples as “blocks” in order to distinguish their properties from those of the core.

[22] Samples of the core and each of the blocks were extracted for various physical, hydraulic and electrical measurements, as described in detail below. An attempt was made to sample a sufficiently wide range of sandstone in order to examine anticipated contrast in properties. However, a limited number of coarse grained sandstone units were sampled as it proved difficult to drill intact subsample plugs because of the weak cementation and the tendency to fragment.

[23] The following measurements were made on subsamples extracted from the core and blocks (measured properties are available as auxiliary material¹): (1) grain size distribution using a Coulter Counter LS230 laser diffraction system, (2) scanning electron microscope (SEM) imaging and analysis to determine sample composition, (3) mercury injection capillary pressure (MICP) tests (using a Micromeritics AutoPore III unit) on 10 mm diameter intact plug samples to determine pore throat size distributions and total porosity, (4) gas permeability on 20 mm diameter intact plug samples, allowing conversion to hydraulic conductivity estimates following *Bloomfield and Williams* [1995], (5) cation exchange capacity (CEC) by saturating exchange sites with sodium using a sodium acetate solution and determining displaced sodium using atomic absorption, and (6) pore surface area on intact subsamples using nitrogen gas adsorption and calculated using the 5 point BET [*Brunauer et al.*, 1938] method. For gas permeability on 20 mm diameter intact plug samples, plugs were extracted in horizontal and vertical directions to account for anisotropy in the sandstone. For pore surface area on intact subsamples using nitrogen gas adsorption and calculated using the 5 point BET method the values were used with MICP measured total porosity to determine surface area to pore volume [see *Knight and Nur*, 1987] assuming a constant grain density of 2.64 g cm⁻³.

[24] In addition, groundwater extracted from the Eggborough drill hole was analyzed for major ions and a synthetic groundwater was made with similar proportions of Cl⁻, NO₃⁻, SO₄⁻, Na⁺, Mg²⁺, Ca²⁺ (32 mg L⁻¹, 707 mg L⁻¹, 112 mg L⁻¹, 21 mg L⁻¹, 28 mg L⁻¹, 229 mg L⁻¹, respectively) resulting in an electrical conductivity of 0.10 S m⁻¹. The synthetic groundwater was used for saturated plug subsamples for electrical measurements described below.

[25] Electrical measurements were made using a Radic Research SIP Fuchs instrument. We adopted the approach for consolidated samples presented by *Vinegar and Waxman*

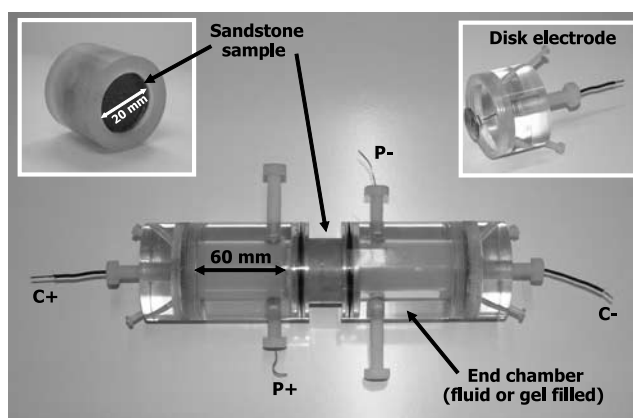


Figure 2. Sample holder arrangement for SIP measurements on sandstone samples. See color version of this figure in the HTML.

[1984] in which end caps, filled with electrically conducting fluid, are used to inject current and measure the potential difference across the sample. Figure 2 shows a photograph of the experimental setup. A few key points are worthy of note.

[26] 1. The 20 mm diameter sandstone sample was sealed in paraffin wax and then encased in acrylic in order to prevent physical degradation of the weakly cemented samples.

[27] 2. Copper disk electrodes were used for current injection, as shown in Figure 2, and current density was maintained to values below 0.05 A m⁻².

[28] 3. Ag-AgCl electrodes were used for potential measurements and inserted into the holder arrangement so that the end of the electrode was flush with the inside of the end cap chamber (i.e., they did not protrude into the current pathway).

[29] 4. Samples were saturated under vacuum with helium degassed synthetic borehole water and left to equilibrate for 24 hours prior to any SIP measurements.

[30] 5. For water saturated sample measurements the end cap reservoirs were filled with the saturating fluid (the synthetic groundwater solution). For unsaturated measurements the end caps were filled with a 4% agar–synthetic groundwater mix preventing ingress of the end cap fluid into the sample by capillary action. Such an approach proved effective for DC resistivity measurements on unsaturated sandstone samples carried out by *Taylor and Barker* [2002].

[31] 6. Saturating fluid and prepared samples were maintained at 20°C in an incubator.

[32] 7. Unsaturated samples were prepared using evaporative drying. We recognize that hysteresis in the electrical response is likely [see, e.g., *Knight*, 1991] but wish, here, to investigate whether any measurable change in SIP behavior is apparent under one desaturating regime. For each saturation level samples were saturated then dried until a target mass loss was reached (using porosity estimated from MICP measurements on plug samples extracted close to the site of the SIP plug sample). In order to minimize the effect of changes in solution salinity we adopted the approach of *Titov et al.* [2004] by using different salinities for different saturation levels, such that the final salinity at the target saturation was assumed equal to the synthetic groundwater.

¹Auxiliary material is available at <ftp://ftp.agu.org/apend/wr/2005WR004202>.

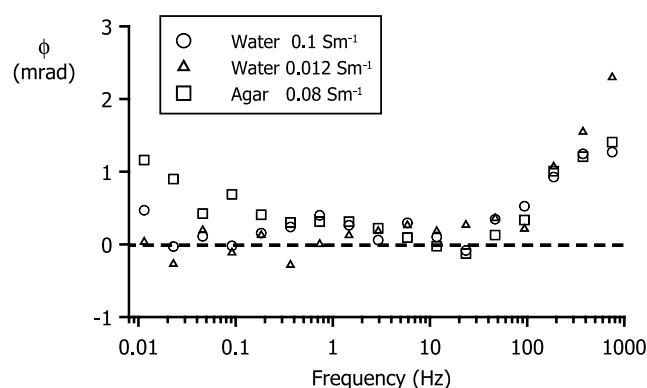


Figure 3. SIP response on “blank” water and Agar samples.

[33] 8. Spectra were measured at 17 frequencies over the range 0.01 Hz to 1 kHz to enable accurate identification of the relaxation curve on the sandstone samples. For each sample, two spectra were measured and an average response recorded for later processing. This allowed removal of occasional obvious outliers.

[34] Tests were carried out on fluid and gel filled sample ‘blanks’ to assess instrument noise. Figure 3 shows phase angle spectra for fluid and Agar filled sample blanks with

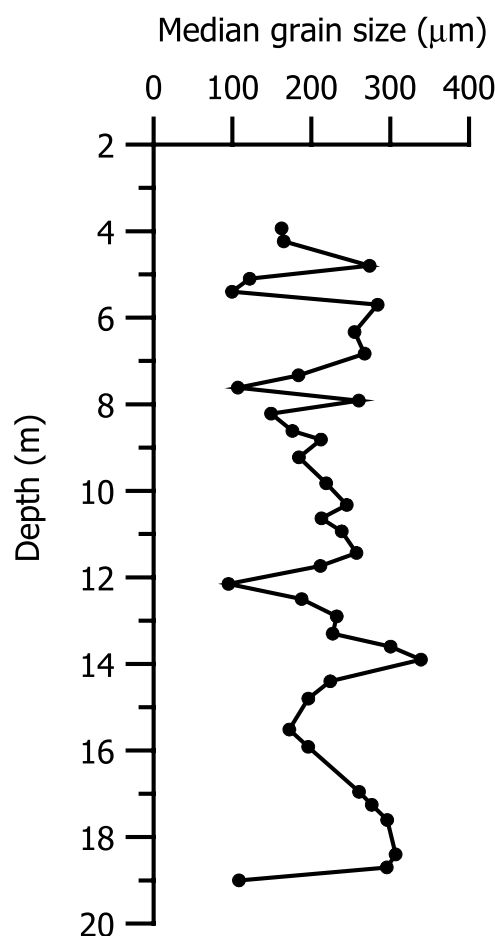


Figure 4. Median grain size distribution in core extracted from the Eggborough site.

electrical conductivities matching those of the samples. The results show capacitive phase angle errors of around 0.5 mrad within the frequency range 0.01 Hz to 100 Hz for the water samples and around 1 mrad for the Agar sample. These errors are less than 5% of the expected phase angles in sandstones samples of the type considered here [Scott and Barker, 2003; Clennell and de Lima, 2003]. Above 100 Hz, capacitive coupling within the measurement system becomes more significant, with errors around 3 mrad for more resistive samples.

3. Physical-Hydraulic Characteristics

[35] To illustrate the contrasting grain size within the core, Figure 4 shows the median grain size (d_{50}) variation with depth. Bands of fine sandstone within the core are clearly visible, as is the apparent fining upward sequences observed in similar cores extracted from the two sites [see, e.g., Binley *et al.*, 2001]. The impact of this layering on vertical flow through the vadose zone at the site has been demonstrated by Cassiani and Binley [2005]. For comparison, d_{50} measurements of 159, 106 and 135 μm were recorded for the three Hatfield sandstone quarry blocks and 92 and 235 μm for the two Eggborough quarry blocks. On the basis of particle size analysis, clay contents (particle size $<2 \mu\text{m}$) varied from 2 to 4%.

[36] A number of subsamples were prepared for Scanning Electron Microscope (SEM) investigation. Analysis of images and spectra from an energy dispersive spectrometer indicated similar principal composition: quartz, feldspars (3 to 7%), haematite (3 to 7%), other minerals including clays, principally kaolinite (3 to 7%). Clay contents estimated from SEM analysis are thus comparable to those based on particle size analysis.

[37] Permeability tests revealed approximately three orders of magnitude range in hydraulic conductivity (see Table 1). Minimal anisotropy was evident for most cases where pairs of samples were extracted in horizontal and vertical directions, giving an average anisotropy of 2.2 (horizontal to vertical).

[38] Following Bloomfield *et al.* [2001], MICP injection pressure data were analyzed by fitting the van Genuchten [1980] (van Genuchten) model:

$$S_{eff} = \frac{\theta - \theta_r}{\theta_s - \theta_r} = \left[1 + \left(\frac{\psi}{\psi_0} \right)^n \right]^{\frac{1-n}{n}}, \quad (7)$$

Table 1. Summary of Physical Properties Obtained From All Core and Block Sandstone Samples

Property	Number of Samples	Minimum	Maximum	Median
Horizontal hydraulic conductivity, m d^{-1}	18	0.034	5.976	0.291
Vertical hydraulic conductivity, m d^{-1}	22	0.014	7.953	0.302
Porosity, %	19	25.8	34.5	29.9
Pore throat size, D_0 , μm	19	12.2	42.3	19.6
van Genuchten model exponent, n	19	1.70	2.55	1.91
CEC, $\text{meq } 100 \text{ g}^{-1}$	22	4.15	8.30	6.22
Surface area per unit pore volume, S_{por} , μm^{-1}	17	11.1	50.7	31.4

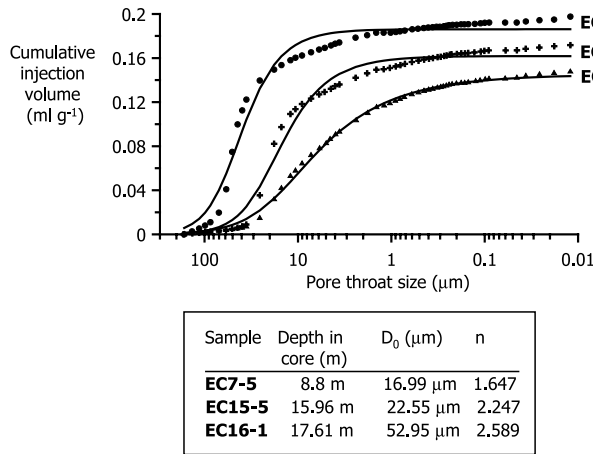


Figure 5. Pore throat distributions for three samples extracted from core together with fitted van Genuchten parameters D_0 and n . Symbols show measured distributions, and lines indicate fitted van Genuchten model.

where S_{eff} is the effective saturation, θ is the wetting fluid content, θ_r is the residual fluid content, θ_s is the fluid content at saturation, ψ_0 and n are parameters.

[39] Pressure values were converted to equivalent pore throat diameter values, D , using the Washburn equation [see Bloomfield et al., 2001]:

$$D = 4\varsigma \frac{\cos \gamma}{\psi}, \quad (8)$$

where ς is the interfacial tension and γ is the wetting angle. Parameters of an equivalent van Genuchten model (equation (7)) are then D_0 , n and S_r , where D_0 is the pore throat size corresponding to the capillary pressure ψ_0 (from equation (8)) and S_r is the residual fluid saturation. Bloomfield et al. [2001] refer to D_0 as a “dominant pore throat size”, equivalent to a pressure at which there is a maximum intrusion of the nonwetting fluid. To illustrate the variability of measured responses, Figure 5 shows example MICP curves and fitted van Genuchten models for three samples extracted from the sandstone core. Table 1 reports the range of responses observed from 19 subsamples

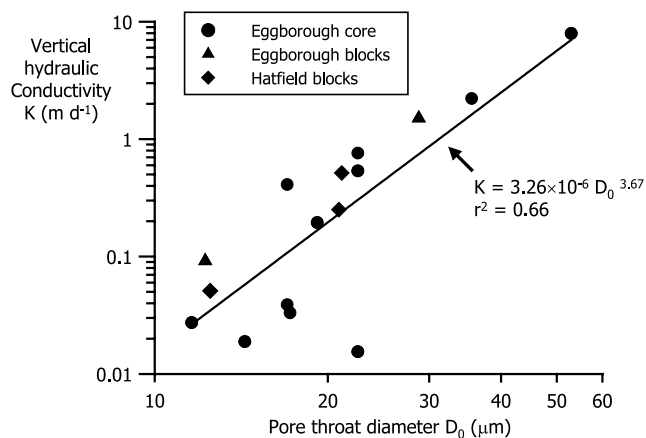


Figure 6. Variation in vertical hydraulic conductivity with characteristic pore throat size, D_0 .

analyzed using MICP (van Genuchten model parameters for each sample are available in the auxiliary material).

[40] The range of measurements of CEC and S_{por} are shown in Table 1. As expected, the two measurements show significant positive linear correlation ($r^2 = 0.69$, graph not shown). Pore throat size, D_0 is significantly correlated with $\log K$, as can be seen in Figure 6. Note that the exponent of D_0 in a power law relation with K is much greater than the value of 2 which is observed in most empirical models [see, e.g., Nelson, 1994]. Similar relationships between D_0 and K to those shown in Figure 6, however, were noted by Bloomfield et al. [2001] for a wide range of UK Triassic sandstone samples.

4. Hydraulic-SIP Relationships in Saturated Sandstone

[41] Example IP spectra (shown in terms of conductivity magnitude and phase angle) at full water saturation for three samples extracted from the drill core are shown in Figure 7. The characteristic pore throat size, D_0 , for the samples are: 52.95 μm (VEC16-1), 22.55 μm (VEC15-5) and 19.16 μm (VEC7-5). These examples reveal ‘classical’ Cole-Cole type relaxation curves with peak phase angles of between 10 and 20 mrad, which is consistent with previously measured spectra in sandstones of this type [Scott and Barker, 2003; Clennell and de Lima, 2003]. The examples in Figure 7 show a measurable increase in phase angle with characteristic pore throat size within the frequency range normally used for field measurements (0.1–1 Hz) indicating that it may be possible to differentiate lithologies, at least, within these sandstones with field IP measurements (electrical parameters for each sample are available in the auxiliary material).

[42] The spectra show a noticeable difference in the frequency at which the peak phase angle occurs, indicating a difference in relaxation time (from equation (4)). This is shown clearly in Figure 8, where a significant (approximately linear) correlation between D_0 and τ is evident. Note also from Figure 8 that no apparent (or measurable) dependence of IP spectra on orientation of sample exists, al-

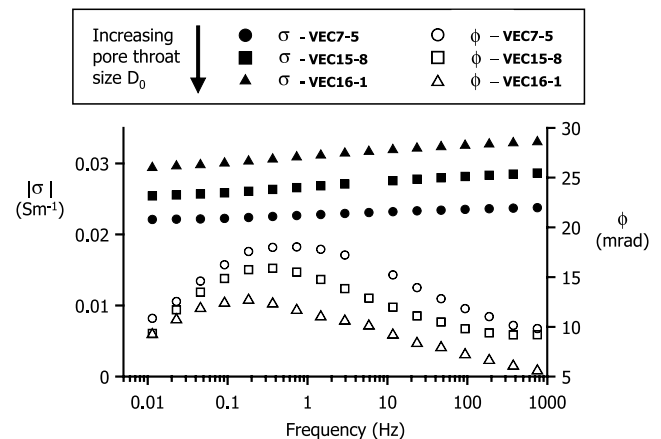


Figure 7. Example SIP response of three samples extracted from the core. Characteristic pore throat size, D_0 , for the samples are 52.95 μm (VEC16-1), 22.55 μm (VEC15-8), and 19.16 μm (VEC7-5).

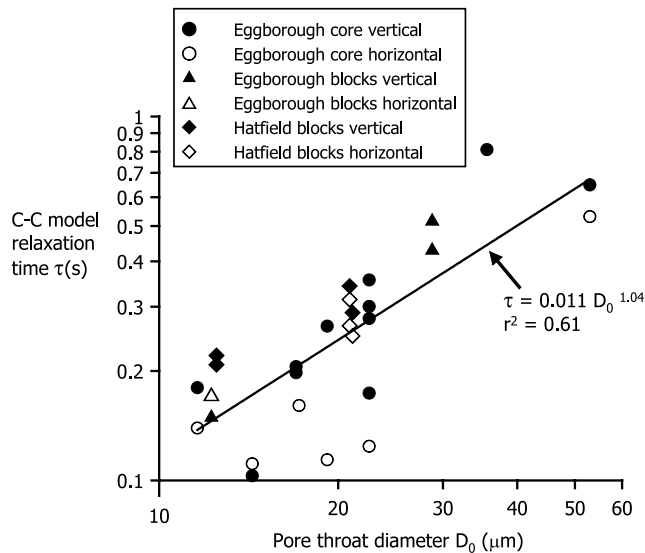


Figure 8. Variation in C-C model relaxation time, τ , with van Genuchten model characteristic pore throat size, D_0 .

though, as stated earlier, hydraulic anisotropy in the samples is relatively weak.

[43] To evaluate the potential use of the Börner *et al.* [1996] IP- K model it is first necessary to test the applicability of the modified Kozeny-Carman relationship in equation (6). Figure 9 shows that, for measurements on the sandstones here, a significant relationship between S_{por} and KF exists. Interestingly the exponent b in equation (6) derived from our measurements (3.6) is close to the theoretically derived value of 3.1 resulting from the PARIS model of Pape *et al.* [1982].

[44] In order to apply the model of Börner *et al.* [1996] we must first demonstrate the link between IP and S_{por} as reported by Börner and Schön [1991]. However, the spectral IP response of our samples clearly does not conform to a constant phase angle model (see Figure 7) and because of the variation in relaxation times it would appear inappropriate to use a single frequency measure of imaginary

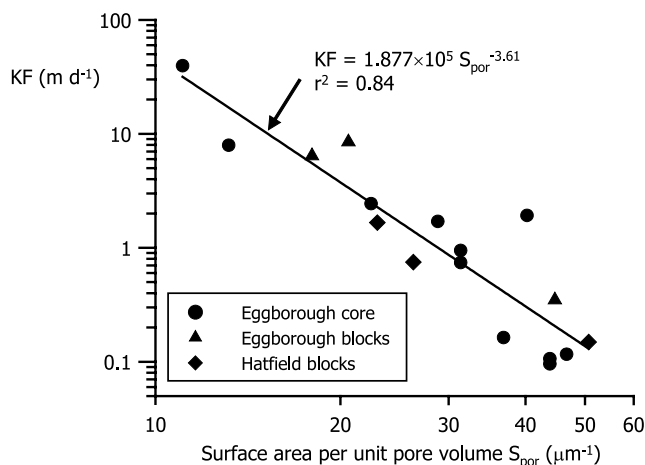


Figure 9. Variation in the product of vertical hydraulic conductivity, K , and the formation factor, F , with surface area per unit pore volume, S_{por} .

electrical conductivity as representative of the IP spectra. Figure 10a shows that the relationship between σ'' and S_{por} is not linear, which demonstrates that the Börner and Schön [1991] relationship indeed does not hold for our samples (using, here, 1.4 Hz as an arbitrary reference frequency, although similar behaviour is seen for other frequencies). While there is a general trend of increasing imaginary conductivity with surface area to pore volume, the relationship is weak ($r^2 = 0.18$) and clearly of limited practical predictive value. In contrast, a clear relationship $\tau \propto S_{por}^{-1}$ exists, as shown in Figure 10b. This dependence is to be anticipated based on the inverse relationship between specific surface area and grain/pore size of a rock. As surface area increases the relaxation time reduces due to an increased contribution of a greater number of smaller grain (or pore throat) sizes that make up the rock. It would appear therefore that the time constant, τ , of these samples provides a measure of some effective grain (or pore) length scale, l , that is, $\tau \propto l$. Comparing τ with a median grain size d_{50} supports this by the relationship $\tau \propto d_{50}$ ($r^2 = 0.62$, graph

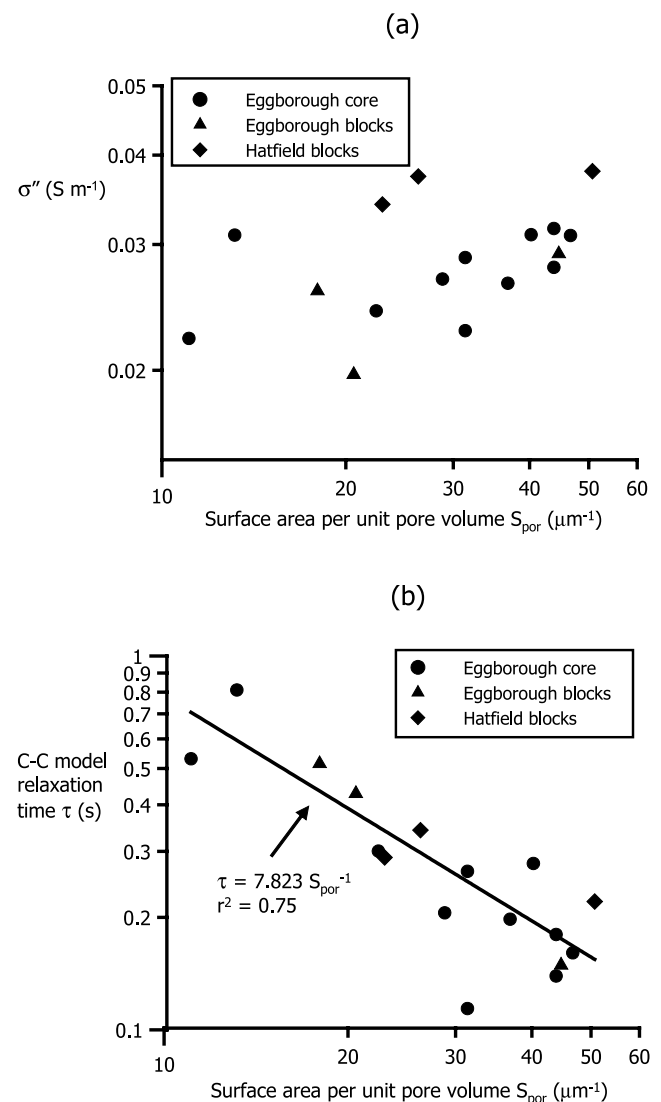


Figure 10. Variation in IP parameters with surface area per unit pore volume, S_{por} . (a) Imaginary electrical conductivity at 1.4 Hz, σ'' ; (b) C-C model relaxation time, τ .

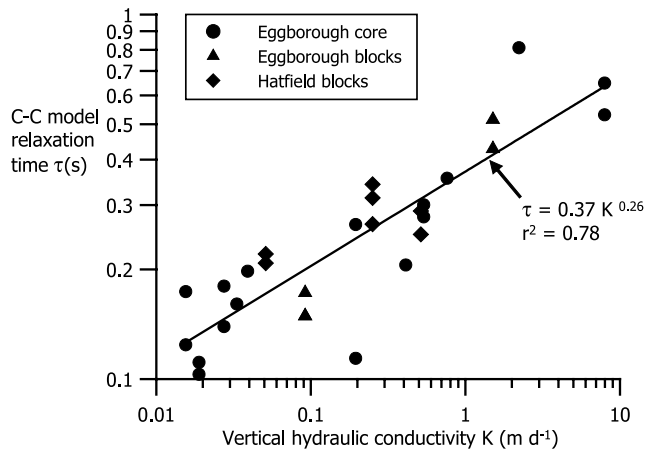


Figure 11. Variation in C-C model relaxation time, τ , with vertical hydraulic conductivity, K .

not shown). These findings are consistent with other reported studies [e.g., Vanhala, 1997b] although it is often assumed that $\tau \propto l^2$ (see discussion by Kemna [2000]). Since we have already established that hydraulic conductivity is proportional to the surface area to pore volume ratio (Figure 9) then a link between relaxation time, τ , and hydraulic conductivity follows, which is demonstrated in Figure 11.

[45] The potential value of this τ - K relationship is significant. For our samples, which do not conform to the constant phase angle assumption necessary to apply previously developed IP – K models, it would appear that variation in hydraulic conductivity is strongly correlated with the Cole-Cole relaxation time, suggesting that this is a much more appropriate measure of IP for these sediments for hydrological characterization. If we consider the τ as a measure of the characteristic size of these features then a link between spectral IP and hydraulic conductivity follows.

[46] We recognize that the C-C model relaxation time is not an intrinsic property of the sample. It is implicitly linked to the conductivity of the sample, as demonstrated by Vanhala [1997b] who summarized findings from a number of earlier studies. Vanhala [1997b] shows evidence of a decrease in τ with increasing electrical conductivity, σ , in saturated samples. Pelton *et al.* [1978] also noted that for an equivalent electrical circuit representing the C-C model, $\tau \propto \sigma_{DC}^{-1/c}$, where σ_{DC} is the DC conductivity and c is the C-C exponent (see equation (3)). As c typically lies in the range 0.1 to 0.6 then again it is expected that τ will decrease as σ increases. For our samples saturated with the same pore fluid we observed a weak relationship between τ and σ_{DC} ($r^2 = 0.17$, graph not shown) suggesting that the electrical conductivity may have only minimal influence on the τ - K relationship.

5. SIP Behavior in Unsaturated Sandstone

[47] Evidence of the effect of water saturation on polarization exists in the literature [e.g., Ulrich and Slater, 2004; Titov *et al.*, 2004], however, we are unaware of previous studies into the impact of saturation on relaxation

times. If the spectral IP relaxation curve is strongly influenced by the distribution of pore throat sizes in saturated samples then it would be logical to conclude that as the fluid saturation is reduced the relaxation curve will change to reflect the size distribution of the pore throats that remain saturated. If we associate τ with an effective length scale associated with hydraulically conductive pores then as saturation decreases a reduction in relaxation time is expected. However, as noted above, the change in conductivity of the sample with saturation will also influence the relaxation time.

[48] Changes in phase spectra, for the samples tested here, are illustrated in Figure 12 for two samples of contrasting pore throat size distributions. In both cases as fluid saturation decreases the relaxation curve is shifted to higher frequencies. If we consider the relaxation curve as a distribution of polarization elements (as postulated by

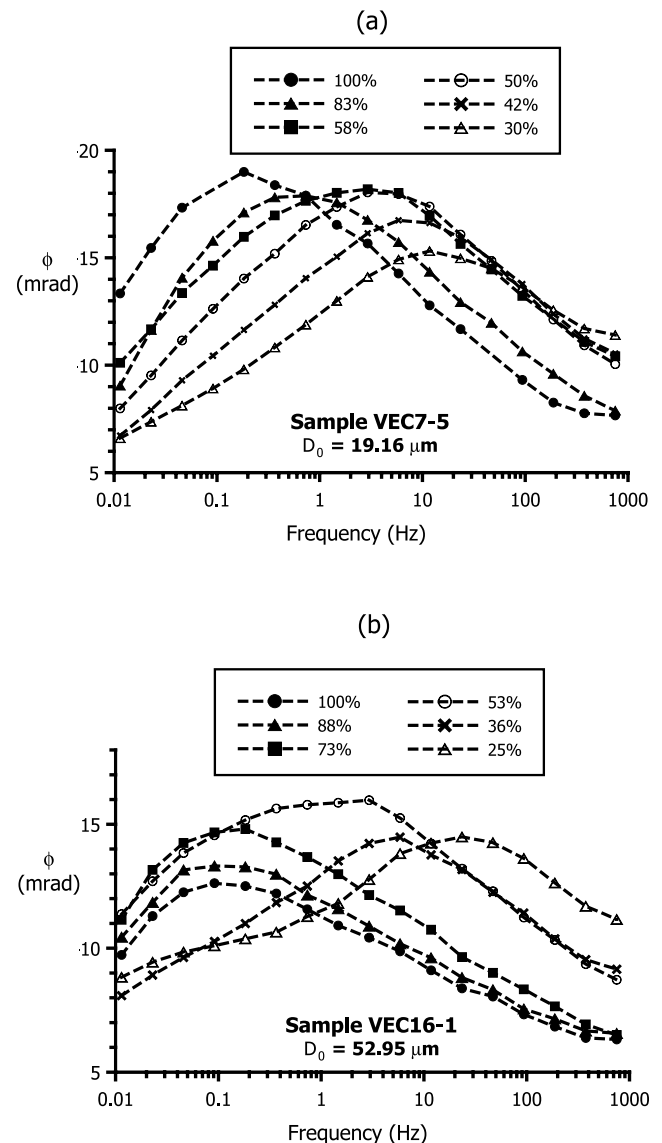


Figure 12. SIP phase angle spectra for core samples under different water saturations. (a) Sample VEC7-5; (b) sample VEC16-1.

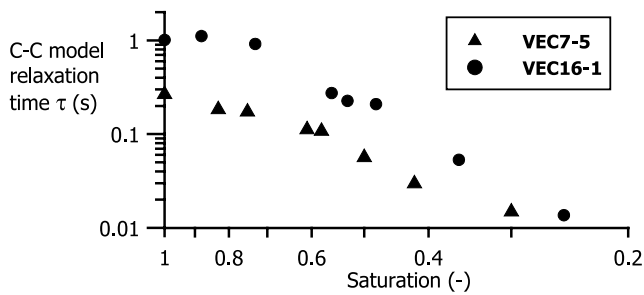


Figure 13. Variation in C-C model relaxation time, τ , with saturation for core samples VEC7-5 and VEC16-1.

Lesmes and Morgan [2001]) then one interpretation is that the smaller pore throats tend to dominate the distribution of relaxation times as larger pores dewater and cause a shift in bulk effective relaxation time (the C-C model parameter τ). Figure 13 shows the variation in τ as a function of saturation for the two samples VEC7-5 and VEC16-1. If we consider the $\tau \propto \sigma_{DC}^{-1/c}$ relationship of Pelton *et al.* [1978], from our experiments c was typically about 0.3 and thus we expect a significant increase in relaxation time due to the conductivity decrease associated with reducing saturation. It would appear therefore that any changes in relaxation time associated with decreased bulk conductivity at lower saturations are masked by what we interpret as effective pore length scale reduction as larger pores dewater.

[49] In saturated samples we observed a one order of magnitude variation in time constant for three orders of magnitude variation in hydraulic conductivity (Figure 11). Results such as those in Figure 13 show one to two orders of magnitude change in the C-C model time constant between saturations of 100 and 25%. This has clear implications on how useful SIP measurements are if they are conducted in the vadose zone. An appropriate correction of field data collected in the unsaturated zone should be applied to reduce to the equivalent response of a fully saturated medium, linked to saturated hydraulic conductivity. The necessary estimate of in situ water saturation could be derived from DC conductivity changes through Archie's law.

6. Implications for Groundwater Model Parameterization

[50] Previously proposed IP- K relationships based on single frequency measures of polarization appear inappropriate for the sandstone under investigation. The results here suggest that where significant phase angle dispersion is observed it is essential to determine the relaxation time as an indicator of hydraulic length scale within the porous medium. Failure to do this would result in meaningless estimates of K if one were to adopt previously developed approaches based on the IP magnitude from a constant phase model. Correlation between the Cole-Cole time constant and hydraulic conductivity was observed and, given the three orders of magnitude variation in observed sample hydraulic conductivity, the potential value of this electrical-hydraulic relationship in groundwater modeling is significant. Our results suggest that it is possible from

SIP alone to determine the variability in hydraulic conductivity within the sandstone, albeit with a site specific relationship.

6.1. Potential Application to Field Studies

[51] A significant value of this observed relationship is the potential to utilize geophysical logging or imaging techniques in the field to establish in situ hydraulic properties. One advantage of such an approach is the fact that geophysical measurements can be made in the field at a variety of measurement support volumes: borehole logging can provide measurements over length scales of centimeters whereas surface or borehole deployed imaging allows measurement length scales of several meters. The observed correlation, however, may have limited practical value as measurements of conductivity spectra over a broad frequency range is difficult, if not impossible in the field. Two major factors limit the frequency range measurable in field studies: (1) measurements at frequencies above 10 to 50 Hz may suffer from effects such as inductive and capacitive coupling, (2) low-frequency measurements are time consuming (one measurement at 0.01 Hz will take ten or more minutes, thus limiting application of subsurface imaging techniques). It is likely that coupling effects can be minimized by the use of, for example, optical cables for sensor connection but this will add considerable cost and complexity to sensor design and may be prohibitively expensive for sacrificial down-hole electrode deployment. It is therefore necessary to assess the value of limited frequency range measurements in predicting hydraulic properties from complex conductivity data.

[52] Clearly, multiple frequency data are required in order to capture the variation in relaxation times exhibited by the samples. However, to make application of practical use we must consider a small number of frequencies, which must also be less than 10 Hz, say, to minimize coupling effects. If we consider phase angles at three frequencies, a quadratic function can be fit through the data and an inflection point in the quadratic function computed. Upon examination of typical spectra (Figure 7, for example) we may capture the shape of the phase spectra over the range 0.1 to 10 Hz. To explore this, phase angle measurements at the frequencies 0.09 Hz, 1.4 Hz and 11.7 Hz were extracted for each sample spectra. A parabolic function of the form:

$$\phi = a(\log f)^2 + b \log f + c, \quad (9)$$

was fit to the three data sets to determine parameters a , b and c are constants, where f is the current injection frequency and ϕ is the measured phase angle. The inflection point on the frequency axis, which we denote here as f_3 , was then computed. Figure 14 illustrates how well this simply derived value correlates with the measured hydraulic conductivity ($r^2 = 0.75$). The fit is poor for high ($>1 \text{ m d}^{-1}$) hydraulic conductivity values since relaxation times will be relatively high and the inflection point not adequately bracketed by the narrow range of frequencies used. We also recognize that such a relationship may have limited predictive capability but it may serve as a useful lithological discriminator for field application.

6.2. Unsaturated Hydraulic Characteristics

[53] In addition to establishing, empirically, a link between saturated hydraulic conductivity and SIP relaxation

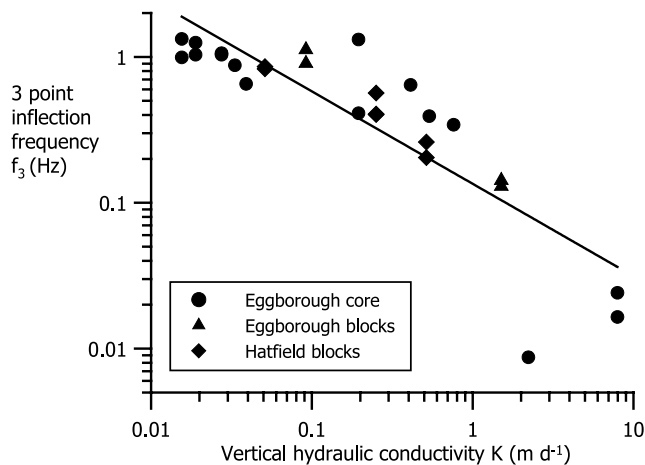


Figure 14. Variation in three point inflection frequency, f_3 , with vertical hydraulic conductivity, K . Solid line shows power law fit $f_3 = 0.135 K^{-0.633}$ ($r^2 = 0.75$).

time, our experiments have confirmed the potential link between a dominant pore throat size and the relaxation time (Figure 8) as previously inferred from observations by *Scott and Barker* [2003]. Such a relationship has potential value for groundwater contamination studies as pore throat size can have a significant impact on, for example, the mobilization of microbiota [see, e.g., *Bloomfield et al.*, 2001]. Perhaps of more significance here, however, is the potential link between unsaturated hydraulic characteristics and SIP parameters. The characteristic pore throat size, D_0 , is related, through the Washburn equation (equation (8)) to the parameter ψ_0 of the van Genuchten pressure-saturation model and thus we have demonstrated an empirical link between the Cole-Cole time constant and this hydrological parameter (although we recognize that such a link may not exist for all soils). In fact, one can argue a parallelism between the van Genuchten and C-C models. Equations (3) and (7) are of a similar form and the main governing parameters of the van Genuchten and C-C models are related to the same soil characteristics in terms of pore (or grain) size. In the van Genuchten model, ψ_0 is a characteristic capillary pressure, linked to an effective pore (or grain) size and the exponent n is associated with the pore (or grain) size distribution.

[54] For the samples tested here we observed C-C and van Genuchten model exponents over a relatively narrow range ($0.25 < c < 0.39$; $1.6 < n < 2.6$) and consequently have insufficient data to test this link empirically. We hypothesize an inverse relationship between these two parameters; as the van Genuchten model exponent n increases for samples with a wider range of pore throat sizes, the dispersion in the IP spectra also increases (c reduces) reflecting the greater contributing range of pore (or grain) sizes.

[55] Our results have demonstrated a significant effect of fluid saturation on conductivity phase spectra, indicating a measurable decrease in relaxation time with reduced saturation. If the relaxation time is implicitly linked to some measure of a length scale influencing hydraulic conductivity then one may hypothesize a relationship between hydraulic conductivity variation with saturation and the τ saturation function. Such concepts require mechanistic

models to support experimental observations, and if developed and confirmed may offer significant potential for characterizing unsaturated hydraulic conductivity through electrical geophysical measurements.

7. Conclusions

[56] Previous field studies at the Hatfield and Eggborough sites have noted the impact of layers of fine sandstones/siltstones on the migration of natural and artificially induced wetting fronts within the predominantly medium sandstone in the vadose zone. The results here suggest that SIP is clearly able to differentiate these lithological types within the sandstones. Utilization of previously proposed models linking IP parameters to hydraulic conductivity appears inappropriate due to the significant dispersion observed in the electrical conductivity phase spectra. Adopting the Cole-Cole model as a macroscopic representation of the IP spectra our results have demonstrated a clear link between the relaxation time of this model and some measure of hydraulic length scale. The relaxation time shows significant correlation with hydraulic conductivity spanning three orders of magnitude, suggesting that from spectral IP data alone, estimates of hydraulic conductivity are possible. We also observed correlation between a characteristic pore throat size and the Cole-Cole relaxation time constant, indicating that it may be possible to predict this parameter of a pressure-saturation model with electrical data.

[57] Hydraulic conductivity estimates are clearly valuable for saturated and unsaturated flow models, although the principle focus of our work is to develop methodologies for characterization of hydraulic properties in the vadose zone. Consequently, experiments were carried out to assess the impact of the IP spectra on changes in fluid saturation. Our results revealed significant impact of saturation on the measured spectra. These results may limit the applicability of hydraulic-electrical models utilizing SIP measurements. In contrast, however, they may offer new opportunities for development of physically based models linking unsaturated hydraulic characteristics with spectral IP data. If we consider the relaxation time as some measure of a length scale influencing hydraulic conductivity then it is logical to propose that this may be extended to unsaturated media and offer a means of determining unsaturated hydraulic conductivity through electrical geophysical measurements.

[58] Recent advances in SIP instrumentation and modeling algorithms, combined with a better understanding of the physical significance of the IP response, encourages field-scale engineering applications of the method. Instrumentation and software now exist, making it possible to obtain 2-D or 3-D images of the polarizability of the subsurface. IP measurements, like DC resistivity measurements, can be taken in cross-hole imaging configurations [*Kemna et al.*, 2004], thus providing information at measurement support volumes which are comparable with those required for hydrological models. We believe that, subject to instrument developments, field-based SIP measurements may allow hydraulic parameterization rather than simply conventional lithological zonation using geophysics. The main future challenge, however, remains the develop-

ment of appropriate mechanistic models which support the link between SIP and hydraulic properties.

[59] **Acknowledgments.** This project would not have been possible without support from many others. John Bloomfield (BGS, Wallingford, UK) provided access to MICP and gas permeability instrumentation; Darren Gooddy (BGS, Wallingford, UK) provided fitting routines for the van Genuchten model; Andreas Kemna (ICG4, Jülich, Germany) provided a Cole-Cole fitting; Yuxin Wu (Rutgers–State University of New Jersey, New Jersey, USA) provided Nitrogen BET analysis; Geoff Johnston (Lancaster University, UK) provided expert technical support for sample holder design and fabrication. Giorgio Cassiani acknowledges partial support from the Italian Ministry of Education, University and Research (MIUR) FIRB grant RBAU01TAL5. The experimental work and analysis was funded under the UK Natural Environment Research Council (NERC) grant NER/A/S/2001/01175. We also acknowledge the two anonymous reviewers, reviewer K. Titov, Associate Editor S. Hubbard, and Editor S. Tyler for their constructive comments that have helped improve the manuscript.

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